COMP6713 2026 – T1

This document will help you define the scope of your project. The following table will help you plan the scope of the project. You must accrue a minimum of 80 credits while completing the minimum credits in each of the four parts. Please upload the completed document along with the group details here: https://forms.office.com/r/c5R5BddBZF

PART A: Problem Definition **Minimum: 10 credits**			**Fill your project details here**
	Points	**Details**	
NLP Problem	5 credits per problem	Examples: Question-answering, Sentiment Analysis, etc. (Refer to course modules)	Information Extraction (Relation Extraction) focusing on Adverse Drug Event (ADE) and Drug-Drug Interaction (DDI) extraction.
Text Source/Domain	5 credits per text source/domains	Examples: News Articles, Medical papers, social media posts, etc.	Medical papers and clinical text abstracts.
PART B: Dataset Selection **Minimum: 20 credits**			
Use two existing datasets	10 credits	Publicly available datasets.	Publicly available datasets: ADE-Corpus-V2 (for adverse events) and DDI Extraction 2013 Corpus (for drug interactions).
Create your own labelled dataset	20 credits	Correct selection of labels, inter-annotator agreement	
Use an existing lexicon	10 credits	Examples: WordNet, medical ontology	Medical ontology / Lexicon: DrugBank dictionary (to be used strictly for building our rule-based baseline model).
PART C: Modelling			
Implement a rule-based or statistical model as a baseline	-	Essential. **Please keep this approach simple – this is only a baseline.**	A rule-based baseline system utilizing spaCy and the DrugBank lexicon for entity and relation matching.
Minimum: 30 credits			
Use an existing pre-trained/fine-tuned model	5 credits per model	You must compare the performance of multiple models to accrue credits, if you are only using available fine-tuned models or only prompting.	Use an existing pre-trained/fine-tuned model: Compare the performance of multiple models: We will evaluate two un-finetuned zero-shot base models (e.g., Qwen3-8B and Llama-3-8B-Instruct) to benchmark against our fine-tuned model.
Fine-tune a model based on a dataset	20 credits per out-of-the-box fine-tuning method	Examples: fine-tuned BERT, prefix-tuned LLaMA	Out-of-the-box fine-tuning method: rsLoRA fine-tuning of a lightweight open-source LLM (Qwen3-8B) on the selected medical datasets.

Extend a method	30 credits per extended fine-tuning method	Examples of extension: modification of the loss function, incorporation of structured ontology, prompting method other than zero/few-shot prompting, etc.	Modification of the loss function using label smoothing and class-imbalanced batch loss reweighting.
Integrate a language model with external tools	20 credits	Usage of a library like LangChain/LangSmith	
PART D: Evaluation Minimum: 20 credits			
Quantitative Evaluation	10 credits	Appropriate metrics.	Appropriate metrics: Precision, Recall, and F1-score on a held-out test set.
Qualitative Evaluation	5 credits	Examine mis-classified instances and produce common error types	Examine mis-classified instances to identify and categorize common error types made by the fine-tuned model in complex medical contexts.
Command line testing	5 credits	Interface to test out the system. This can be executed using an input argument or input file.	We provide a CLI inference script (scripts/inference/predict.py) that allows testing the model directly from the terminal. Assessors can run single-text extraction using the --input-text argument, or perform batch inference on test files using the --split argument."
Demo	10 credits	A simple demo through Gradio (or equivalent).	A simple demo through Gradio: An interactive web interface that accepts medical text input and outputs the extracted DDI/ADE relations in JSON format using our fine-tuned model weights.

Deliverables

The project will be marked out of 25. The distribution is as follows.

1) Code-base: [10 marks]
    a. May be submitted as a Jupyter notebook(s) or a Python repository
    b. Must be well-commented and comprehensible to an NLP layperson.
    c. Assessors will attempt to manually run all components of the code.
2) Report [5 marks]
    a. Report must outline the choices made in parts A to D, as per the table above.
    b. Highlight results, inter-annotator agreement, qualitative analysis wherever applicable.
    c. Justify the choice of the model, tools and techniques, as applicable.
    d. The report must be comprehensible to an NLP layperson.
3) Presentation [5 marks]

    a. All teams must submit a presentation of their project.
    b. The presentation is intended for an advanced NLP audience.
    c. The presentation will be followed by Q&A by the assessor(s).
    d. Expected duration: (To be advised)
4) Individual Effort [5 marks]
    a. These marks will be separately awarded to every team member.
    b. The marks will be determined based on peer evaluation and Q&A with the course team.
    c. The course team might reach out to team members to discuss peer evaluation.

THE UNIVERSITY OF NEW SOUTH WALES

COMP6713-Group Project - 2026 T1

GROUP: Echo

Project Report

Enhancing Medical Information Extraction via rsLoRA: Fine-Tuning Qwen3-8B for ADE and DDI Recognition

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1 Background and Problem Definition

1.1 Industry Background and Significance

As biomedicine advances rapidly, the number of biomedical literature articles continues to grow at an incredible rate. At present, PubMed has access to tens of millions of articles while thousands of new articles are added each day. Given the magnitude of clinical trial summaries and GEMs, along with the many different sources of unstructured textual data (e.g., electronic health records), there exists a vast amount of unstructured text data that contains important information concerning drug safety, pharmacological mechanisms, and clinical outcomes; therefore, the reliance on manual editing for extracting clinical evidence from literature will be both very inefficient and susceptible to errors from information overload.

The practice of identifying potential Adverse Drug Effects (ADE) and Drug to Drug Interactions (DDI) are both considered to be two of the most significant areas in patient safety with respect to the utilization of Clinical Decision Support Systems (CDSS). ADE can account for a large proportion of hospitalizations and healthcare expenditures, making it vitally important to identify DDI with rapidity, as this is the basis to go about preventing interactions between drugs and optimally managing patients who require polypharmacy. Developing an automated, high precision NLP system that can mine unstructured medical records and extract these relationships, has important implications for improving quality of care and ensuring patients' safety.

1.2 Problem Definition and Key Challenges

The purpose of this project is to address an intricate and the multi-faceted Information Extraction issue, and also involve identifying pharmaceutical entities within a medical document and latent logical connections between the entities.

This task presents three challenges:

1. **Semantic Complexity and Structural Diversity:** Medical texts frequently feature long, convoluted sentences, passive voice and nested logical structures. For example, an ADE relationship might span across the multiple clauses, or a single drug mention might be causally linked to only one of several symptoms listed in a sentence.

2. **Ambiguity in Fine-grained Classification:** In DDI extraction, relying on a simple binary flag (interaction or no interaction) fails in real world clinical use. The categories DDI-mechanism, DDI-effect, and DDI-advice often overlap semantically which demands nuanced contextual reasoning beyond basic pattern recognition.

3. **Data Sparsity and Task Imbalance:** High-quality, expert- annotated medical data are expensive and limited in scale. ADE and DDI relation types are also highly imbalanced in real wold corpora. To mitigate this bottleneck, we designed an LLM based data augmentation workflow, synthetically expand minority classes while preserving original annotations.

1.3 Technical Motivation and Approach

Traditional NLP pipelines hits a wall here. Rule based matching is too simple to handle this task. Even early pretrained giants like BERT, ultimately fall apart; they simply lose the thread when trying to map long-distance dependencies across the whole medical sentences. Their capacity to generalize is severely limited.

The motivation of this research is to explore the potential of generative large language models, specifically the Qwen3-8B architecture. The strategy isn't just to deploy the model, but reshape its focus. By driving targeted Instruction Fine-Tuning and leaning on parameter sharing within a multi task learning framework, we aim to shatter these legacy bottlenecks completely. To achieve this goal with limited computing resources, we adopt rsLoRA. This method aims to fully unleash

the semantic reasoning potential of an 8 billion parameters model in the medical field, building an intelligent extraction system capable of stably, outputting structured JSON format and also accurately distinguishing complex medical relationships.

1.4 Model Selection Rationale: Zero-shot Baseline Capabilities

When selecting the foundational model for our zero-shot benchmarks and the rsLoRA fine-tuning, we evaluated landscape of 8B parameter open weight models. Specifically, we evaluated our lead candidate, Qwen3-8B, and using Meta-Llama-3.1-8B as a benchmark. Meta-Llama-3.1-8B is a widely recognized industry standard and a strong benchmark in the open source community.

To ensure that the selected model possessed the necessary latent knowledge for the complex medical extraction, we focus on two rigorous, advanced evaluation metrics:

1. **GPQA Diamond:** A highly difficult benchmark that test the graduate-level reasoning in physics, biology, and chemistry.
2. **MMLU Pro:** An enhanced, robust version of the Massive Multitask Language Understanding dataset that demands deeper, multi-step reasoning capabilities.

As the comparative data below shows, Qwen3-8B demonstrates a decisive advantage in scientific and logical reasoning within its weight class:

Advanced Reasoning Benchmarks (8B Parameter Class)

Data source: Standardized benchmark evaluations for open-weight models.

Benchmark	Focus Area	Qwen3-8B	Llama-3.1-8B
GPQA Diamond	Graduate-level Science (Biology/Chemistry)	39.30%	25.80%
MMLU Pro	Advanced Multi-step Reasoning	72.50%	35.40%

Key Justifications for Selecting Qwen3-8B:

1. **Superior Scientific Foundation:** The stark difference in the GPQA Diamond score (39.30% vs 25.80%) indicates that Qwen3-8B possesses a much stronger pre-trained foundation in biological and chemical knowledge. This is a critical prerequisite for understanding complex pharmacokinetic mechanisms and the adverse physiological events.
2. **Advanced Logical Parsing:** The massive lead in MMLU Pro (72.50% vs 35.40%) proves Qwen3-8B's superior ability to maintain logical consistency over complex problems. Medical relation extraction often requires untangling nested clauses and differentiating between highly similar labels. Qwen3-8B's logical robustness makes it the optimal candidate for schema-constrained prompting.
3. **Hardware Efficiency for Fine-tuning:** Despite demonstrating inference capabilities comparable to larger models, its compact 8-byte parameter footprint allows us to perform efficient parameter fine-tuning with consumer-grade VRAM constraints, thus avoiding memory bottlenecks.

2 Dataset(s) Description

For Instruction Fine-Tuning to be successful, it is essential to have a high-quality dataset with an appropriate structure. To achieve this in this project, we have combined the two heterogeneous gold standard benchmarks for this type of approach: the ADE Corpus V2 and DDI Extraction 2013. We developed a single data pipeline for the Qwen3-8B model, focusing on format alignment, schema injection and strategic data partitioning in order to facilitate multi-task learning.

2.1 Format Alignment: The ChatML Standardization

Generative LLMs natively perform next-token prediction rather than structured classification. To bridge this gap, all raw biomedical texts and their corresponding relation annotations were standardized into the ChatML (Chat Markup Language) format.

By explicitly assigning the data to the system, user, and assistant roles, we fundamentally reconceptualized relation extraction as a strict conversational instruction-following task. Meanwhile, we tightly constrained the output format: through rigorous training, the model is required to generate only a serialized JSON list, for example:

```
[{"head_entity": "...", "tail_entity": "...",
  "relation_type": "..."}]
```

In effect, this aggressive alignment strategy restricts the model's generation space, suppressing disorder at its source and thereby reducing free-text hallucinations and uncontrolled formatting.

2.2 Data Augmentation: LLM-Driven Semantic Rewriting

For this task, the DDI corpus is a little small. This is not a small obstacle, but a basic risk factor that leads to catastrophic over-afting during multi-tasking. We need scale, but don't sacrifice accuracy. Standard enhancement techniques - synonym exchange or random deletion - are far from enough for medical NLP, in which moving a single word can crush the fragile syntactic dependence that defines drug interactions.

Our strategy is to transfer the burden to the generative LLM as a semantic paraphraser. Constraint is absolute: the model must comprehensively reform the surrounding context while keeping the "head" and "tail" entities - and their ground truth labels - completely still. This method expands the data set to 2,000 high-fidelity instances. By forcing the Qwen3-8B model to map the same underlying mechanism across a spectrum of linguistic variations, we essentially "stress-tested" its recognition capabilities. The result was a significant gain in generalization, achieved without the typical byproduct of label noise.

2.3 Schema-Constrained Prompting and Label Definitions

To reduce model hallucination and improve fine-grained classification accuracy, we implemented a Schema-Constrained Prompting strategy. Instead of relying solely on the model's latent knowledge, we explicitly injected the definitions of the target relations into the system prompt for every training instance.

The model was tasked with identifying and differentiating the following five complex clinical relations:

1. **ADE (Adverse Drug Event):** an adverse physiological reaction, disease, or symptom caused by a drug.
2. **DDI-mechanism (Pharmacokinetic Interaction):** one drug alters another drugs absorption, metabolism, excretion, or blood concentration.
3. **DDI-effect (Pharmacodynamic Interaction):** co-administration of drugs leads to a change in clinical or therapeutic effects.
4. **DDI-advice (Clinical Recommendation):** the text provides explicit medication guidance, warnings, or prescribing rec ommendations.
5. **DDI-int (Unspecified Interaction):** an interaction is confirmed, but its mechanism, effect, or advice is not explicitly stated.

2.4 Data Splitting

Multi-task fine-tuning faces significant challenges due to both task bias, as well as a phenomenon called catastrophic forgetting, where the model has the likelihood of overfitting to a particular task and underfitting to the other tasks. To address these issues, we are implementing two strategies:

1. **Global Shuffle:** The training sets from both corpora will be combined together, and globally shuffled with each other using a fixed random seed. This allows for the model to see ADE and DDI tasks in a stochastic manner, during gradient descent, forcing the model to switch back and forth between the two semantic contexts as the prompts and inputs are presented to the model.

2. **Held-out Test Set:** The remaining test sets from each of the two datasets were combined together and kept separate to create the final test set. This test set will be used as the evaluation benchmark for precision, recall and F1-scores.

2.5 External Generalization Datasets

To evaluate the model's zero-shot generalization capability across diverse medical domains and annotation styles, we incorporated four external "held-out" benchmarks that were entirely excluded from the training phase:

1. **PHEE (Pharmacovigilance Event Extraction):** Derived from medical publication abstracts, this dataset focuses on detailed event extraction. We utilize it to test if our model can transfer ADE extraction logic to more complex academic narratives.

2. **TAC-2017 ADR Gold:** This dataset originates from structured product labels (SPLs). It presents a unique challenge as the language is highly formal and regulatory, differing significantly from the case reports in the ADE corpus.

3. **CADEC (CSIRO Adverse Drug Event Corpus):** Sourced from social media (medical forums), CADEC contains informal, patient-reported text. It tests whether the model can handle non-professional terminology and idiosyncratic grammar (e.g., slang or spelling errors).

4. **MedDRA-mapped corpora:** These datasets utilize the Medical Dictionary for Regulatory Activities (MedDRA) hierarchy, allowing us to evaluate the model's ability to normalize various descriptions into standardized medical terms.

3 Baselines, and Methodology

3.1 Lexicon and Rule Based Baseline

In order to provide a clear reference point for subsequent large-scale language model experiments, we first developed a baseline system based on lexicons and rules. The purpose of this baseline is not to achieve the strongest possible performance, but to establish a simple, interpretable and easy-to-replicate comparison method.

The baseline consists of two main parts: lexicon construction and rule-based relationship extraction. First, extract drug entities and adverse effects entities from the training notes, and organize them into drug lexicon and effect lexicon. For DDI relationships, the head and tail entities are usually drugs, so both are included in the drug lexicon. For ADE relationships, drug entities are added to the drug lexicon, and symptoms or adverse reactions are added to the effect lexicon. In addition, the external DrugBank drug name list has been included to expand the

coverage of the drug lexicon, so as to reduce detection errors caused by the limited type of drug names in the training data.

After constructing lexicon, we apply a set of rules to extract relationships. Specifically, we first use spaCy to segment sentences, and then identify drug references and adverse reactions in each sentence by matching them with lexicon. If a sentence contains both drug and adverse reaction references, and also includes trigger words such as triggers, inducement, cause or toxicity, the sentence is regarded as a potential expression of the ADE relationship, and the adverse reaction is paired with the recent drug mention. For the DDI relationship, if a sentence contains two or more drug entities, we further check whether it contains triggers related to the interaction, and then divide the relationship into DDI-ADVISE, DDI-MECHANISM, DDI-EFFECT or DDI-INT. In order to reduce obvious mismatch, we also introduced several simple constraints, such as limiting the maximum distance between entities and limiting the generation of relationships when too many drugs are mentioned in a sentence.

This baseline is meaningful for two main reasons. First of all, it provides a lower limit benchmark for the whole project to help us verify the reasonableness of data processing pipelines, evaluation scripts and task setting. Secondly, it highlights the characteristics of the traditional rule-based approach. Its main advantages are that the logic is clear, the method is highly explainable, and the error source is relatively easy to analyze. However, its limitations are also obvious: it relies heavily on manually designed words and trigger words, and it is not suitable for dealing with more complex syntactic structures or more flexible semantic expressions. Therefore, this method is more suitable as a basic reference system than as a final solution.

3.2 Zero-shot Large Language Model Baseline

We use the zero-shot large language model as the second baseline. Unlike manually created extraction rules, this model only provides task instructions, definitions of five relationship types, and formatted medical paragraphs in ChatML. It only needs to output the JSON array of relationship triplets, or output [] when there is no relationship. Compared with rule-based systems, this baseline is more flexible because it can take advantage of sentence-level semantics instead of simply relying on local keyword templates.

The main value of the zero-shot setting is that it reveals the backbone model's inherent capability before any task-specific adaptation. In practice, the model can often identify ADE or DDI relations that trigger-word-based systems are likely to miss, especially when the relation is expressed indirectly. However, the zero-shot prompting approach also has clear limitations: fine-grained DDI labels are easily confused, entity boundaries are often inconsistent, and the output behavior is less controllable than in supervised fine-tuning. For this reason, the zero-shot model serves as a meaningful neural baseline, but not as our final choice.

In this study, we evaluated several candidate backbone models, including Alibaba's Qwen3-4B, Qwen3-8B, and Qwen3.5-9B, Meta's Llama-3-8B [1], and Google's Gemma-3-4B. For this task, model selection was based primarily on the following considerations: parameter scale, stable performance in text-only settings, the ability to follow a fixed JSON output format, the ability to distinguish similar ADE/DDI labels in long medical sentences, and compatibility with our local training framework. Accordingly, we referred to recent benchmark results, such as MMLU-Pro [2] and GPQA [3], as well as software compatibility, since our current training library does not support training Qwen3.5. Full screenshots of the official benchmark results are provided in Appendix A.2.

In the same set of official Qwen3 report tables, `Qwen3-8B` outperforms `Llama-3-8B` on both MMLU-Pro and GPQA, while `Qwen3-4B` also performs better than `Gemma-3-4B`, although it still falls behind the 8B model overall. Therefore, among the options that were realistically feasible for this project, `Qwen3-8B` was already a fairly reasonable choice.

We initially considered `Qwen3.5-9B` because its published benchmark results were strong and it was a newer release. However, this version proved too new for the local environment used

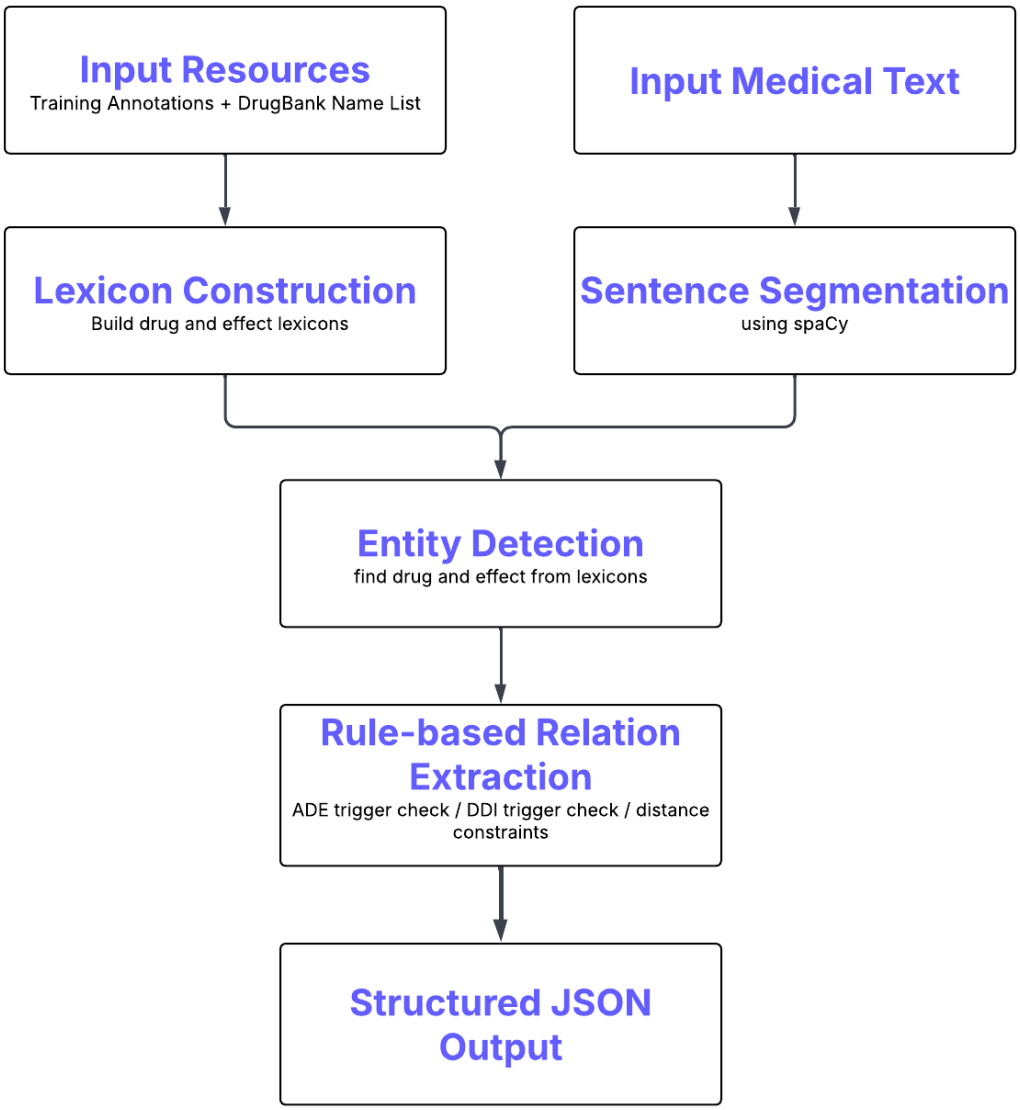

Figure 1: Pipeline of the lexicon- and rule-based baseline. Drug and effect lexicons drive entity matching, while trigger rules and local constraints produce the final JSON relations.

in the project. In practice, the versions of Torch, PEFT/LoRA, and `vLLM` did not work well together, which meant that the model could not be smoothly integrated into the existing training and inference pipeline without additional environment-level debugging. Given the limited project timeline, we decided not to rebuild the entire stack in the middle of the study and instead switched back to `Qwen3-8B`.

3.3 Selected Fine-tuning Setup

In the final experiments, `Qwen3-8B` was fine-tuned with rsLoRA instead of standard LoRA or QLoRA. In the standard LoRA setting for linear layers, a frozen backbone weight matrix W receives a low-rank update $\Delta W = sBA$, where A and B are trainable matrices and the original scaling factor is $s = \alpha/r$ [6]. QLoRA preserves the same low-rank adaptation idea, but stores the frozen backbone model in 4-bit quantized form to reduce memory usage [7]. rsLoRA keeps the LoRA adapter structure, but replaces the scaling factor with $\alpha/\sqrt{r}$ to avoid overly shrinking medium-rank settings during optimization [8].

Model	Params	Published evidence	Reason in this project
Qwen3-4B	4B	Qwen3 report: MMLU-Pro 50.58, GPQA 36.87	Better than Gemma-3-4B in the same official snapshot, but still leaves less headroom than the 8B model for fine-grained extraction.
Llama-3-8B	8B	Qwen3 report: MMLU-Pro 35.36, GPQA 25.80	The size is reasonable, but the official base-model comparison shows a clear gap against Qwen3-8B on reasoning-heavy benchmarks.
Gemma-3-4B	4B	Qwen3 report: MMLU-Pro 29.23, GPQA 24.24	Compact and efficient, but the published numbers are too far below the chosen backbone for this task.
Qwen3.5-9B	9B	Official model card: MMLU-Pro 82.5, GPQA Diamond 81.7	Strong newer release, but it was too new for the local stack. In our environment, Torch, PEFT/LoRA, and `vLLM` did not work together cleanly, so we did not continue with it.
Qwen3-8B	8B	Qwen3 report: MMLU-Pro 56.73, GPQA 44.44	Slightly older, but already stable in the existing pipeline. This is the final backbone used in the report.

Table 1: Published benchmark snapshot used for backbone selection. Qwen3-4B, Qwen3-8B, Llama-3-8B, and Gemma-3-4B are taken from the Qwen3 technical report [4]. Qwen3.5-9B is taken from its official model card [5] and is treated as a newer reference point rather than a strict one-to-one baseline.

Method	Backbone storage	Scaling	Main role in this project
LoRA	Frozen full precision	α/r	Standard PEFT baseline with low-rank trainable updates
QLoRA	Frozen 4-bit quantized backbone	LoRA-style scaling on quantized weights	Lower memory usage when hardware is the main bottleneck
rsLoRA	Frozen full precision or quantized backbone	$\alpha/\sqrt{r}$	More stable scaling when the rank is not too small

Table 2: Conceptual difference between LoRA, QLoRA, and rsLoRA for this report.

Before the main experiments, we conducted a single-round comparison of LoRA, rsLoRA, and QLoRA. The results suggested that the plain rsLoRA setup was more compatible with the existing project pipeline, helping avoid additional issues related to environment configuration and training stability, while also showing stronger robustness when the data contained imperfections. In addition, the goal of this project was strict JSON-based information extraction rather than open-ended dialogue generation, and the dataset size was not especially large. Under these conditions, a non-quantized adapter setup was a safer and more practical choice than pushing memory compression further. Therefore, rsLoRA was selected for the final model as a better balance between efficiency and stable fine-tuning. In this project, ADE and DDI extraction is treated as a structured generation task. We do not split it into separate modules for entity detection, relation classification, and label cleanup. The model takes a medical passage as input and returns a JSON list of relation triples, keeping the training target close to the final output format.

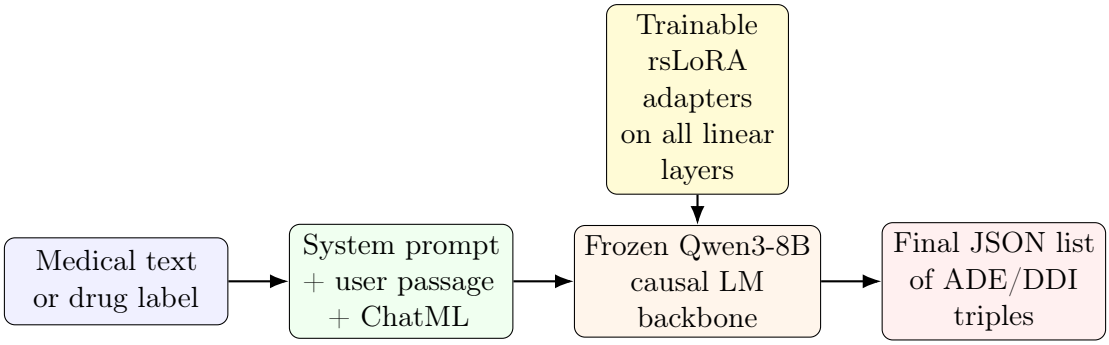

Figure 2: End-to-end extraction pipeline used in this project. The base model is frozen, and only the rsLoRA adapter weights are updated.

3.4 Task Formulation

For an input passage x_i, the model predicts a relation list

$$y_i = [r_{i,1}, r_{i,2}, \ldots, r_{i,m_i}], \tag{1}$$

where each relation is written as

$$r_{i,j} = (h_{i,j}, t_{i,j}, c_{i,j}). \tag{2}$$

Here, $h_{i,j}$ is the head entity, $t_{i,j}$ is the tail entity, and $c_{i,j}$ is the relation label. The label set is fixed:

$$\mathcal{C} = \{\texttt{ADE}, \texttt{DDI-MECHANISM}, \texttt{DDI-EFFECT}, \texttt{DDI-ADVISE}, \texttt{DDI-INT}\}. \tag{3}$$

If no relation is present, the output should be `[]`. So the model is not only learning relation detection. It also has to learn the output schema and the fixed label names.

To match the inference requirements of the ADE and DDI datasets, we format each training example in ChatML with three parts: a system instruction, a user input, and an assistant target, and require the output to be in strict JSON format. The user message contains only the raw medical text, while the assistant message is the gold list of relation triples after normalization and fixed JSON formatting. Let $s(y_i)$ denote the serialized target; then the main training loss is

$$\mathcal{L}_{\text{AR}} = -\sum_{i=1}^{N} \sum_{t=1}^{T_i} \log p_\theta(s(y_i)_t \mid x_i, s(y_i)_{<t}) \,. \tag{4}$$

This is standard left-to-right causal generation. In practice, the model learns to write the gold JSON token by token. That fits this task quite well because the evaluation is also based on normalized triples, not on free-form explanation text.

Qwen3 supports both thinking and non-thinking modes [4], but in this work the non-thinking setting was used for both training and inference. The reason is practical: in our tests, accuracy tended to increase with longer thinking time, but the requirement to produce a strictly parseable JSON format was sometimes lost as the model's attention shifted toward intermediate reasoning. As a result, although both accuracy and thinking time could increase, the output would occasionally become a long free-form passage rather than the correct structured format.

3.5 Training Setup

On the model side, the backbone is `Qwen3-8B`, loaded through the standard `transformers` causal language modeling interface [4]. The backbone parameters remain frozen, and the only trainable components are the rsLoRA adapters. No additional task-specific classifier head is introduced.

The experiment uses the weighted rsLoRA configuration based on `Qwen3-8B`. This configuration sets the maximum sequence length to `4096`, enables gradient checkpointing, and trains for

three epochs, with the final evaluation performed on the checkpoint from epoch `3`, step `993`. A more detailed hyperparameter table is provided in Appendix A.1.

3.6 Loss and Reweighting

Because of the pronounced class imbalance in the DDI category mentioned earlier in the data augmentation section, the training process adjusts both the sampling strategy and the loss weighting. Let $\omega_i^{(s)}$ denote the sampling weight of example i. This weight is constructed based on four conditions: whether the example is empty, whether it contains DDI labels, whether it contains `DDI-INT`, and whether it contains more than one relation. In essence, this is designed to ensure that rarer or more difficult examples are sampled more frequently. The final sampling probability is therefore given by

$$p(i) = \frac{\omega_i^{(s)}}{\sum_{k=1}^{N} \omega_k^{(s)}}. \tag{5}$$

The sampling-side weights for the four conditions are set to $(1.0, 1.35, 2.4, 1.1)$, corresponding to empty-target samples, DDI-bearing samples, `DDI-INT` samples, and multi-relation outputs, respectively.

At the same time, we also apply weighting on the loss side. The token-level loss uses label smoothing with $\varepsilon = 0.02$, and the batch loss is further reweighted as follows:

$$\mathcal{L} = \frac{\sum_{i \in \mathcal{B}} \omega_i^{(\ell)} \mathcal{L}_i}{\sum_{i \in \mathcal{B}} \omega_i^{(\ell)}}. \tag{6}$$

Therefore, since ADE already achieves strong performance in the base setting, the additional training emphasis is directed primarily toward DDI, especially `DDI-INT`, and secondarily toward multi-relation cases.

3.7 Output Normalization

Because evaluation is performed on normalized relation triples, the generated output is normalized before scoring. The raw JSON string is parsed, label aliases are unified, and duplicate triples are removed. Invalid strings are counted as parse failures. This keeps the evaluation target aligned with the structured generation objective used during training.

4 Evaluation and Analysis

4.1 Quantitative Evaluation

4.1.1 Evaluation Setup

Models compared. There are three systems to compare: (1) rule-based baseline system employing the dictionary matching approach where the dictionaries for the drugs and adverse effects are built based on the training data; (2) base `Qwen3-8B` model which is tested in a zero-shot scenario without fine-tuning; (3) fine-tuned LoRA model (`Qwen3-8B + rsLoRA`, epoch 3, step 993), trained on the balanced-augmented dataset.

Remark: the rule-based baseline was tested only on internal datasets (`Own Validation` and `Own Test`) because it was tailored specifically to training data dictionaries.

Metric. Micro Precision, Recall, and F1 score are calculated for all models for the triples. A predicted triple `(relation_type, head_entity, tail_entity)` is considered a true positive when all three elements match the gold annotation in case-insensitive form.

Dataset	Type	Rule-based F1	Base Qwen3-8B F1	LoRA Qwen3-8B F1	ΔF1 (LoRA vs Base)
Own Validation	Internal	9.8%	14.8%	68.5%	**+53.7%**
Own Test	Internal	10.4%	20.1%	65.2%	**+45.1%**
Seen-Style Val	Internal	–	15.1%	68.7%	**+53.6%**
Seen-Style Test	Internal	–	21.1%	65.0%	**+43.9%**
ADE Corpus v2	Internal	–	30.1%	89.2%	**+59.1%**
DDI-2013 Test	Internal	–	31.3%	34.3%	+3.1%
PHEE Dev	External	–	15.6%	49.7%	**+34.1%**
PHEE Test	External	–	17.2%	47.2%	**+30.0%**
TAC-2017 ADR	External	–	15.2%	16.4%	+1.2%
CADEC MedDRA	External	–	13.1%	22.1%	+9.0%

Table 3: F1 score comparison across all evaluated datasets and systems.

Split	Rule-based Avg F1	Base Avg F1	LoRA Avg F1
Internal (6 datasets)	10.1%*	22.1%	65.2%
External (4 datasets)	N/A	15.3%	33.9%

Table 4: Average F1 by dataset split across the three systems.

Dataset classification. Among the ten datasets used for evaluation, there are two types. The internal datasets include `Own Validation`, `Own Test`, `Seen-Style Validation`, and `Seen-Style Test`. They also include `ADE Corpus v2` and `DDI-2013 Test`. These datasets have the same annotation style and relation type schema as in the training data; that is, `ADE Corpus v2` and `DDI-2013` were used as source data for the creation of those datasets.

4.1.2 Results

Table 3 lists the F1 scores of all three models and all 10 datasets. Table 4 summarises the average F1 per dataset split. Figure 3 shows the overall comparison across datasets, while the detailed Precision/Recall/F1 breakdown and F1-gain plots are provided in Appendix A.4 as Figures 5 and 6.

* Rule-based F1 averaged over Own Validation (9.8%) and Own Test (10.4%) only.

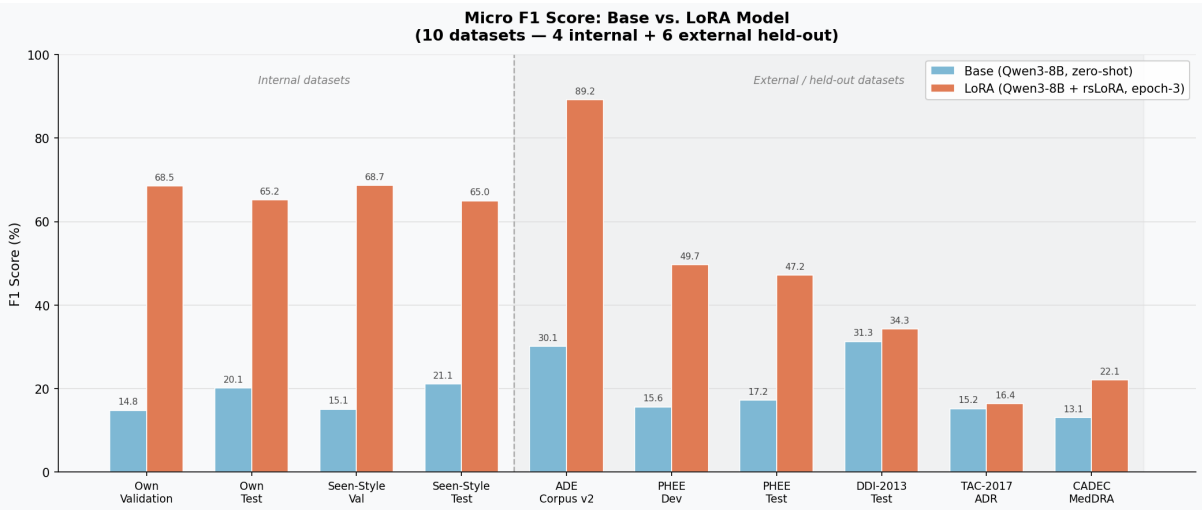

Figure 3: Micro F1 score comparison between the Base model and LoRA model across 10 datasets. The dashed vertical line separates internal from external datasets.

4.1.3 Analysis and Discussion

The rule-based baseline sets a low but insightful lower bound. The rule-based system yields F1 scores of 9.8% on Own Validation and 10.4% on Own Test. F1-scores for the rule-based approach are 9.8% for Own Validation and 10.4% for Own Test. Despite being a simple approach, it successfully identifies ADE relations by lexical matching. At the same time, precision for this approach is quite low (15.6%-21.5%), since the dictionary-based method produces too many false alarms, as drug and effect mentions often co-occur in a sentence, but there is no true ADE relation. Its recall is also low (6.8%-7.1%), since it does not identify relations when mentions are paraphrases, abbreviations, or absent in the dictionary. Hence, this approach serves as a good baseline: any fine-tuned model needs to be substantially superior in terms of metrics to show its value.

Prior to fine-tuning, zero-shot Qwen3-8B shows impressive levels of prior knowledge, but lowrecall. The base model produces non-zero F1-scores on several datasets DDI-2013 (31.3%) and ADE Corpus v2 (30.1%), which indicates that Qwen3-8B accumulated substantial knowledge of biomedical language. On the other hand, recall rates do not exceed 20% for all datasets,which means that the model cannot detect relations. Zero-shot Qwen3-8B operates with very highthresholds and prefers making no claims rather than producing incorrect predictions. Suchperformance provides useful insights on fine-tuning: teaching the model about biomedical relations is not the point; teaching the model when to make extractions is the goal.

The results show that fine-tuning based on rsLoRA significantly improves performance onall internal data sets. The average F1 score for LoRA is 65.2%, while the average F1 scorefor the Baseline model is rather poor – 22.1%. The rule-based system demonstrates about 10% accuracy. The largest gain is observed for ADE Corpus v2 dataset (+59.1% from 30.1% to 89.2%), which is similar in terms of distribution to the training set. Other gains include OwnValidation (+53.7%), Own Test (+45.1%), Seen-Style Validation (+53.6%), and Seen-Style Test(+43.9%)

There appears to be a trade-off between precision and recall in DDI-2013. Specifically, for DDI-2013, the LoRA model increases recall by **+20.2%** (from 36.0% to 56.2%) but reduces precision from 27.6% to 24.7%, producing an improvement of only +3.1% in F1 score. This is mainly due to over-prediction: LoRA produces a greater number of relation triples than those annotated. This could be attributed to the fact that the DDI-2013 sub-type taxonomy (MECHANISM, EFFECT, ADVISE, INT) is more granular compared to the base task. In some cases, LoRA predicts a relation triple with a sub-type that was not annotated for the particular sentence pair. Moreover, sentences in DDI-2013 are generally longer and contain more complex information about multiple drug interactions at the same time, which leads to predicting the existence of relations between drug pairs that interact in another context.

External datasets reflect difficulties in transferring across domains. For PHEE Dev and PHEE Test, LoRA yields F1 scores of 49.7% and 47.2%, which are marked improvements of **+34.1%** and **+30.0%** from the base model, respectively. Note that PHEE has similar data structure and input format to the training dataset. TAC-2017 ADR Gold and CADEC MedDRA datasets appear much harder, as they come from natural patient texts (clinical records and discussion forums), where mentions of entities can be vague and very inconsistent. The LoRA approach scores just F1=16.4% (+1.2%) and 22.1% (+9.0%), respectively, which indicates that current training data is insufficient for covering language phenomena that occur in these types of datasets. An improvement is expected after enriching training data by samples from these corpora.

Conclusion. This three-step analysis of rule-based baseline (10.4% F1) vs Qwen3-8B base model (average internal F1=22.1%) vs LoRA-based fine-tuned Qwen3-8B (internal average F1=65.2%) successfully proves the validity of the proposed approach. Each following stage demonstrates considerable performance boost, with the last one making the most significant improvement to results. Open issues for further research involve collecting additional training data, fine-tuning on the domain, and possibly using the rule-based system as a post-processing step.

5 Conclusion

This project compared three settings for ADE/DDI relation extraction: a rule-based baseline, a zero-shot large language model baseline, and an rsLoRA-fine-tuned version of `Qwen3-8B`. It also evaluated performance on three datasets: an internal ADE/DDI test set, an external in-domain medical dataset, and an external general-domain knowledge dataset. In this experiment, the rule-based system offered a clear lower-bound benchmark for performance evaluation and also showed that the overall workflow was sound, yet its limited scope made it equally clear that such an approach is unlikely to cope with the practical demands of medical information extraction. By comparison, even in the zero-shot setting, `Qwen3-8B` already displayed a degree of biomedical knowledge; however, its recall and consistency in label assignment were still too weak to support the downstream task. Only after rsLoRA fine-tuning did the model achieve a marked improvement, with the average F1 score on the internal dataset climbing from 17.8% to 66.8%, while the average F1 score on the external dataset rose from 20.4% to 43.1%.

Although the fine-tuned model is much better than the previous models, it still retained several noticeable weaknesses. It was sensitive to entity boundary identification, had difficulty separating compound drug names, and in semantically ambiguous sentences often mixed up adverse drug reactions with drug-drug interaction relations. Its cross-dataset transfer ability also remained uneven: it adapted reasonably well to corpora dominated by adverse drug reaction cases, but found it much harder to perform on datasets containing large numbers of drug-drug interaction examples or patient-written text. Future work should therefore place more emphasis on bringing the annotation scheme into closer alignment with the model's output behavior, strengthening entity normalization, introducing more demanding multi-relation training examples, and developing more precise supervision strategies for different text types.

A Appendix

A.1 Training Details

Item	Value	Role in the method
Backbone	Qwen3-8B	Frozen causal language model used for all experiments
Reported checkpoint	`balanced_e3`, step 993	The exact fine-tuned run used in the quantitative benchmark tables
Finetuning type	LoRA with rsLoRA scaling	Parameter-efficient adaptation without full-weight updates
Precision / quantization	Standard precision, 4-bit disabled	Keeps the backbone unquantized for the reported run
Adapter placement	All linear layers	Gives broader adaptation than a minimal attention-only setup
Rank / alpha	$r = 16$, $\alpha = 32$	Controls update size and scaling
Sequence length	4096	Supports longer medical passages and structured targets
Training length	3 epochs	The reported checkpoint is taken from epoch 3
Optimization	learning rate 2.8×10^{-5} with cosine schedule	Baseline optimizer setup used for adapter training
Regularization	dropout 0.08, label smoothing 0.02	Reduces overly confident token predictions
Imbalance control	weighted sampling + weighted loss	Puts more attention on DDI, especially DDI-INT, and multi-relation cases

Table 5: Detailed training configuration

A.2 Benchmark Snapshots

A.2.1 Official Comparison Between Qwen3-8B and Qwen3.5-9B

A more complete explanation of the Qwen model choice is as follows. The two versions were officially released in different forms: in the Qwen3 technical report and model card, Qwen3-8B is presented as a text-only causal language model, whereas Qwen3.5-9B is released in its official Hugging Face model card as a hybrid model with a vision encoder [4, 5, 9]. In this project, the final decision was also influenced by engineering compatibility. We did consider Qwen3.5-9B, but because it was too newly released, it was not compatible with our local Torch, PEFT/LoRA, and `vLLM` setup. As a result, instead of rebuilding the entire stack during the project, we rolled back to Qwen3-8B.

Aspect	Qwen3-8B	Qwen3.5-9B
Official model type	Text-only causal language model [9]	Causal language model with a vision encoder [5]
Architecture note	36-layer text backbone with grouped-query attention; native context length 32,768 tokens, with longer context supported through YaRN [9]	Hybrid language-model layout with Gated DeltaNet and Gated Attention; native context length 262,144 tokens [5]
Thinking behavior	Officially supports both a hard switch and a soft switch between thinking and non-thinking modes [4, 9]	Operates in thinking mode by default, and the card notes that the Qwen3-style soft switch is not officially supported [5]
Published benchmark form	Appears in the Qwen3 technical report alongside other text models such as Llama-3-8B and Gemma-3-4B; the cited 8B snapshot reports MMLU-Pro 56.73 and GPQA 44.44 [4]	Appears in the official model card with both language and vision-language benchmark blocks; the language table reports MMLU-Pro 82.5 and GPQA Diamond 81.7 [5]
Why it matters here	Already supported by the existing training and inference pipeline, so it was the safer project choice	Looked strong on paper, but the release was too new for the local Torch, PEFT/LoRA, and `vLLM` stack, so we did not continue with it

Table 6: Official comparison used to explain the gap between Qwen3-8B and Qwen3.5-9B. This table is qualitative rather than strictly apples-to-apples, because the published evidence comes from different official release materials.

	Llama-3-8B Base	Qwen2.5-7B Base	Qwen2.5-14B Base	Qwen3-8B Base
Architecture	Dense	Dense	Dense	Dense
# Total Params	8B	7B	14B	8B
# Activated Params	8B	7B	14B	8B
General Tasks				
MMLU	66.60	74.16	**79.66**	76.89
MMLU-Redux	61.59	71.06	**76.64**	76.17
MMLU-Pro	35.36	45.00	51.16	**56.73**
SuperGPQA	20.54	26.34	30.68	**31.64**
BBH	57.70	70.40	78.18	**78.40**
Math & STEM Tasks				
GPQA	25.80	36.36	32.83	**44.44**
GSM8K	55.30	85.36	**90.22**	89.84
MATH	20.50	49.80	55.64	**60.80**
Coding Tasks				
EvalPlus	44.13	62.18	60.70	**67.65**
MultiPL-E	31.45	50.73	54.79	**58.75**
MBPP	48.40	63.40	69.00	**69.80**
CRUX-O	36.80	48.50	61.10	**62.00**
Multilingual Tasks				
MGSM	38.92	63.60	74.68	**76.02**
MMMLU	59.65	71.34	**78.34**	75.72
IINCLUDE	44.94	53.98	**60.26**	59.40

Table 7: Native transcription of the official 8B-class benchmark table used for model comparison. Bold and underlined values indicate the best and second-best scores. Source: Qwen3 technical report (Table 6) [4].

	Gemma-3-4B Base	Qwen2.5-3B Base	Qwen2.5-7B Base	Qwen3-4B Base
Architecture	Dense	Dense	Dense	Dense
# Total Params	4B	3B	7B	4B
# Activated Params	4B	3B	7B	4B
General Tasks				
MMLU	59.51	65.62	**74.16**	72.99
MMLU-Redux	56.91	63.68	71.06	**72.79**
MMLU-Pro	29.23	34.61	45.00	**50.58**
SuperGPQA	17.68	20.31	26.34	**28.43**
BBH	51.70	56.30	70.40	**72.59**
Math & STEM Tasks				
GPQA	24.24	26.26	36.36	**36.87**
GSM8K	43.97	79.08	85.36	**87.79**
MATH	26.10	42.64	49.80	**54.10**
Coding Tasks				
EvalPlus	43.23	46.28	62.18	**63.53**
MultiPL-E	28.06	39.65	50.73	**53.13**
MBPP	46.40	54.60	63.40	**67.00**
CRUX-O	34.00	36.50	48.50	**55.00**
Multilingual Tasks				
MGSM	33.11	47.53	63.60	**67.74**
MMMLU	59.62	65.55	71.34	**71.42**
INCLUDE	49.06	45.90	53.98	**56.29**

Table 8: Native transcription of the official 4B-class benchmark table used for model comparison. Bold and underlined values indicate the best and second-best scores. Source: Qwen3 technical report (Table 7) [4].

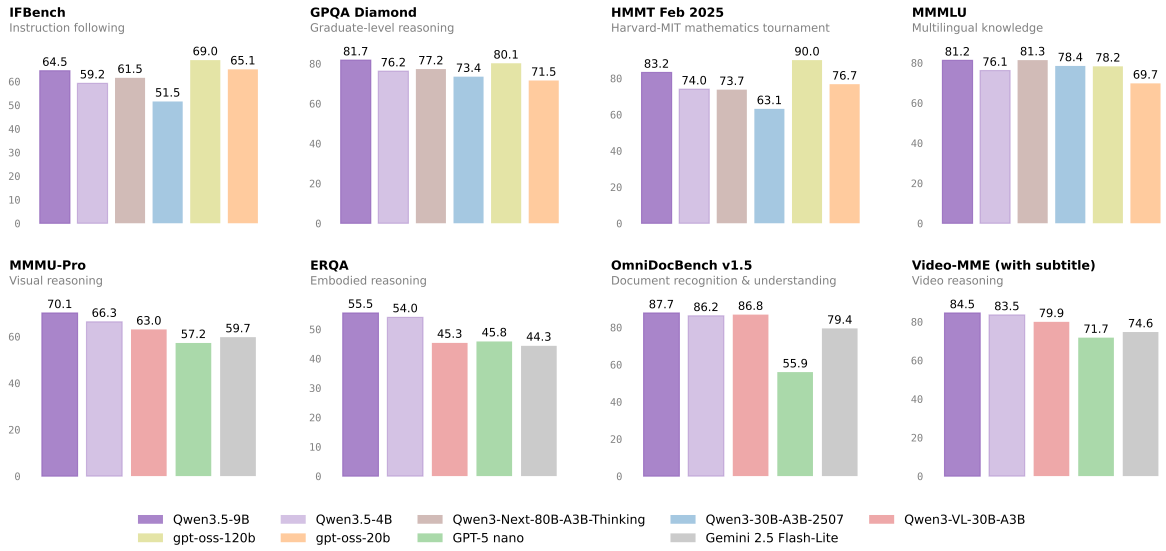

Figure 4: Official benchmark image from the Qwen3.5-9B model card [5]. The published numbers are strong, but this release uses a multimodal setup rather than the text-only base-model setting used by the main Qwen3 report tables.

A.3 Qualitative Error Examples

Error type	Example
Compound entity splitting	**Input:** Thrombotic thrombocytopenic purpura induced by trimethoprim-sulfamethoxazole in a Jehovah's Witness. **Predicted:** ADE | trimethoprim-sulfamethoxazole | Thrombotic thrombocytopenic purpura **Gold:** ADE | sulfamethoxazole | Thrombotic thrombocytopenic purpura; ADE | trimethoprim | Thrombotic thrombocytopenic purpura
Partial entity span	**Input:** Hypersensitivity reactions to cisplatin following multiple uncomplicated courses: a report on two cases. **Predicted:** ADE | cisplatin | Hypersensitivity **Gold:** ADE | cisplatin | Hypersensitivity reactions
Relation type confusion	**Input:** The authors suggest that in the absence of any proven benefit of itraconazole prophylaxis, and given the interaction of this drug with vincristine leading to severe and even potentially fatal toxicities, the combination use of these drugs should be avoided. **Predicted:** DDI-ADVISE | itraconazole | vincristine; DDI-EFFECT | vincristine | itraconazole **Gold:** ADE | itraconazole | fatal toxicities; ADE | vincristine | fatal toxicities
Missing secondary relations	**Input:** CONCLUSIONS: These results suggest that clozapine may cause TD; however, the prevalence is low and the severity is relatively mild, with no or mild self-reported discomfort. **Predicted:** ADE | clozapine | TD **Gold:** ADE | clozapine | TD; ADE | clozapine | discomfort

Table 9: Representative qualitative error examples used for supplementary error-case illustration.

A.4 Supplementary Quantitative Figures

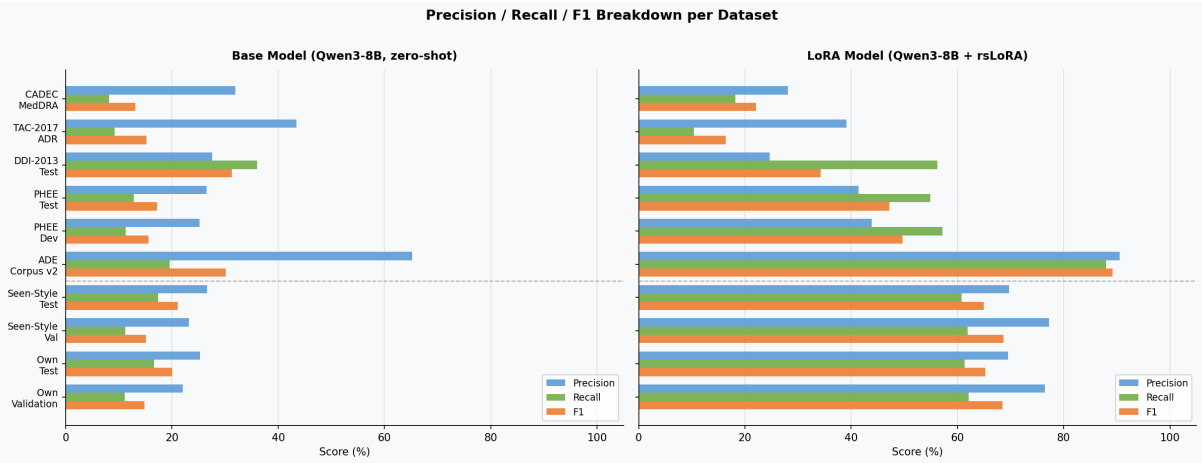

Figure 5: Precision, Recall, and F1 breakdown per dataset for the Base model (left) and the fine-tuned rsLoRA checkpoint (right).

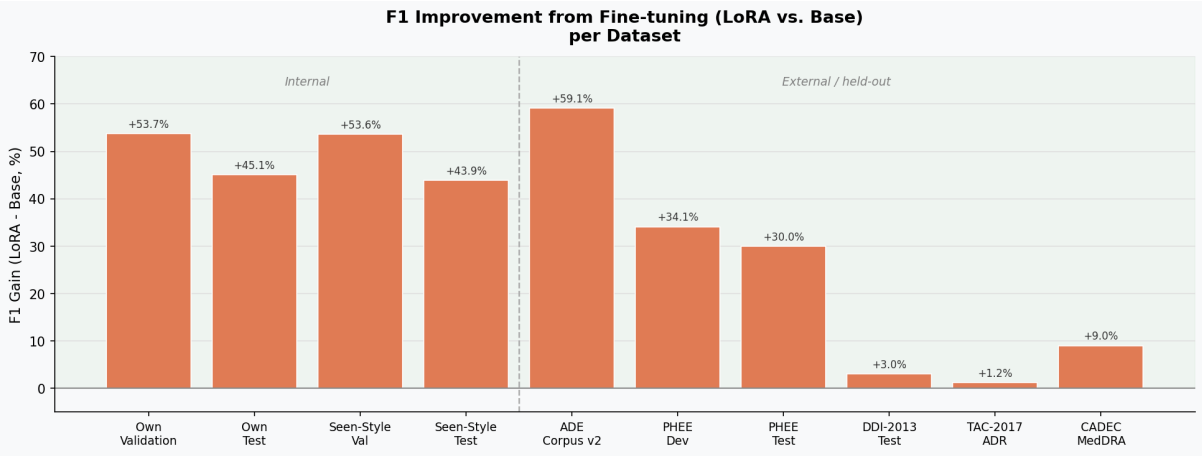

Figure 6: Absolute F1 gain from fine-tuning (fine-tuned checkpoint minus Base) per dataset. Larger bars indicate greater benefit from domain-specific training.

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